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Ideally exact categories

Beyond semiabelian categories, a categorical approach to ideals

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The problem

Let $(R, 1_R)$ be an unital ring, and $I \triangleleft R$ an ideal.

$$R \in \underline{Ring},\ I \in \underline{Ring_*}$$

$$1_R \in I \implies I = R$$

Thus, no proper ideals are unital. Moreover

$$I \text{ ideal of } R \in \underline{Ring} \iff \exists\ f \in \underline{Ring_*} \text{ st } I = \mathrm{Ker}(f)$$

Given a morphism $f \in \underline{Ring}$, $\mathrm{Ker}(f)$ is not defined in $\underline{Ring}$, but it is in $\underline{Ring_*}$. In fact

$$\underline{Ring_*} \text{ pointed} : 0 = 1 = \{0\}$$

$$\underline{Ring} \text{ not pointed} : 0 = \mathbb{Z} \text{ and } 1 = \{0\}$$

This is categorically unnatural!

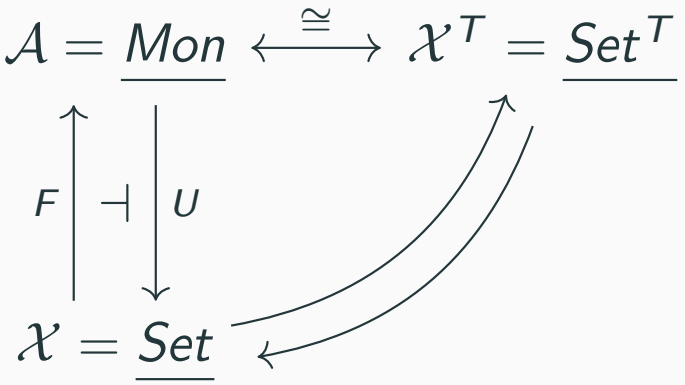
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Preliminaries

Given an adjunction $\mathcal{A} \underset{U}{\overset{F}{\underset{\longrightarrow}{\overset{\longleftarrow}{\bot}}}} \mathcal{X}$, one can consider the associated monad $T = UF : \mathcal{X} \to \mathcal{X}$.

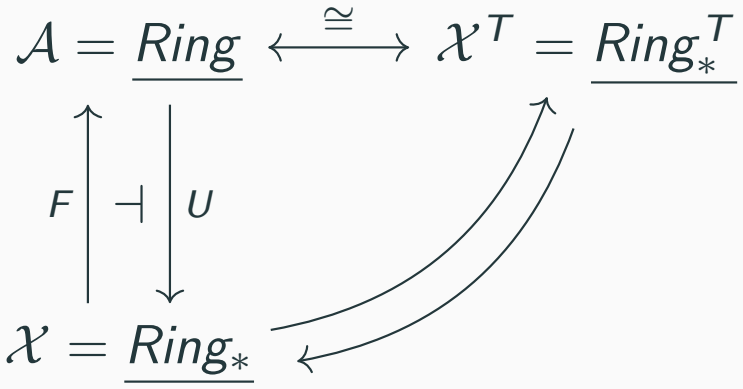
Definition

U is monadic if $\mathcal{A} \cong \mathcal{X}^T$.



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The problem



$$U : \underline{Ring} \to \underline{Ring_*} \qquad F : \underline{Ring_*} \to \underline{Ring}$$
$$R \mapsto \mathbb{Z} \ltimes R$$

Theorem

$U : \underline{Ring} \to \underline{Ring_*}$ is monadic.

Preliminaries

Def

A finitely complete category $\mathcal{A}$ is said to be **exact** if

- Every morphism in $\mathcal{A}$ admits a (regular epi, mono) factorization;
- Regular epis in $\mathcal{A}$ are pullback stable;
- Any equivalence relation is the kernel pair of some arrow.

Example

$\underline{Set}$, $\underline{Grp}$, $\underline{Ring}$, $\underline{Ring_*}$ are exact categories.

Preliminaries

Definition

A category $\mathcal{A}$ is said to be **semiabelian** if

- It is exact;
- It is pointed;
- It is protomodular;
- It has binary coproducts.

Example

$\underline{Grp}$, $\underline{Ring_*}$ are semiabelian, while $\underline{Ring}$ fails to be.

The problem

Normal subobjects in a semiabelian category are the kernels of some morphism.

Theorem

Given an object A in a semiabelian category, it holds that

$$\mathrm{NSub}(A) \overset{\cong}{\longleftrightarrow} \mathrm{Quot}(A)$$

$\underline{Grp} \rightsquigarrow$ Normal subgroups

$\underline{Ring} \rightsquigarrow$ Ideals ?

The problem

$\underline{\mathit{Ring}}$ is not semiabelian, however it's natural to associate

$$\underline{\mathit{Ring}} \rightsquigarrow \text{Ideals}$$

This can be formalized by considering ideals of $\underline{\mathit{Ring}}$ as kernels of morphisms in $\underline{\mathit{Ring}_*}$, which is semiabelian.

$$\underline{\mathit{Ring}} \rightsquigarrow \underline{\mathit{Ring}_*} \rightsquigarrow \text{Ideals}$$

This suggests a generalization of the previous theorem, in a wider class of categories, which we call **ideally exact categories**

$$\mathrm{NSub}\big(U(A)\big) \overset{\cong}{\longleftrightarrow} \mathrm{Quot}(A)$$

with a suitable functor U to a semiabelian category.

Pullback functor

Theorem

Let $p : E \to B$ be a morphism in a category $\mathcal{A}$ with pullbacks.

$$(\mathcal{A} \downarrow B) \underset{p^*}{\overset{p_!}{\underset{\longrightarrow}{\overset{\longleftarrow}{\bot}}}} (\mathcal{A} \downarrow E)$$

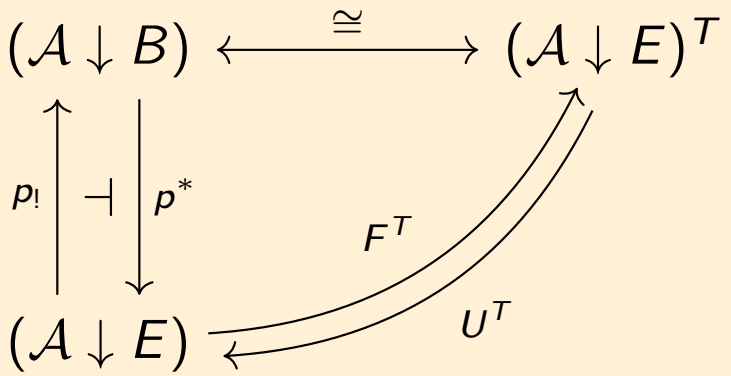
$$\begin{array}{c} A \\ \downarrow{\scriptstyle \alpha} \\ B \end{array} \quad \longmapsto^{p^*} \quad \begin{array}{ccc} E \times_B A & \xrightarrow{proj_2} & A \\ \downarrow{\scriptstyle proj_1} & & \downarrow{\scriptstyle \alpha} \\ E & \xrightarrow{p} & B \end{array}$$

$$\begin{array}{c} D \\ \downarrow{\scriptstyle \delta} \\ E \end{array} \quad \longmapsto^{p_!} \quad \begin{array}{ccc} D & & \\ \downarrow{\scriptstyle \delta} & \searrow^{p\delta} & \\ E & \xrightarrow{p} & B \end{array}$$

Pullback functor

Theorem

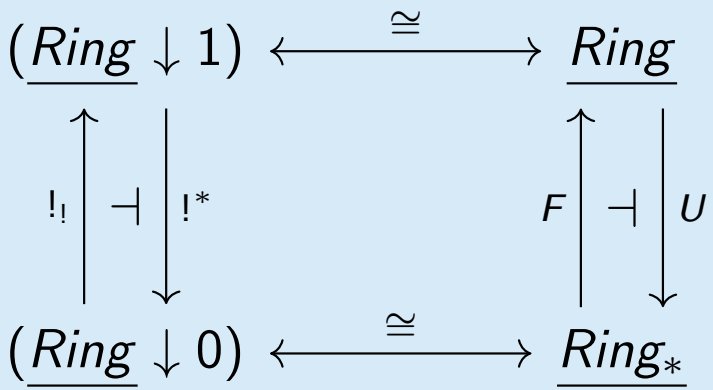
In an exact category, if $p : E \to B$ is a regular epi, then p^* is monadic.



Pullback functor

Example

$$! : 0 \to 1$$



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Lifting results on E-M categories

Theorem

Let $T : \mathcal{X} \to \mathcal{X}$ be a monad

$$\mathcal{X} \text{ has finite limits} \implies \mathcal{X}^T \text{ has finite limits}$$

and U^T creates them.

$$\mathcal{X} \text{ protomodular} \implies \mathcal{X}^T \text{ protomodular}$$

If T preserves regular epimorphisms and kernel pairs, then

$$\mathcal{X} \text{ exact} \implies \mathcal{X}^T \text{ exact}$$

Lifting results on E-M categories

Theorem

Let $T : \mathcal{X} \to \mathcal{X}$ be a monad. If $\mathcal{X}^T$ has reflexive coequalizers, then

$$\mathcal{X} \text{ has finite coproducts} \implies \mathcal{X}^T \text{ has finite coproducts}$$

Theorem

Let $\mathcal{A}$ be an exact and Mal'tsev category. Then $\mathcal{A}$ has reflexive coequalizers.

Lifting results on E-M categories

Example

$$U : \mathcal{A} \to \mathcal{X} \text{ monadic i.e. } \mathcal{A} \cong \mathcal{X}^T$$

$\mathcal{A} \to \mathcal{X}$	$+$ coproduct in $\mathcal{A}$	$+$ coproduct in $\mathcal{X}$
$\underline{Mon} \to \underline{Set}$	$*$ free product	$\sqcup$ disjoint union
$\underline{Ring} \to \underline{Ring_*}$	$*_{\mathbb{Z}}$ free product with amalg	$*$ free product

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Ideally exact categories

Definition

A category $\mathcal{A}$ is said to be **ideally exact** if it satisfies any of the following conditions:

1. – $\mathcal{A}$ is exact, protomodular, has finite coproducts;
   – $! : 0 \to 1$ is a regular epi;
2. – $\mathcal{A}$ is exact;
   – There exists $U : \mathcal{A} \to \mathcal{X}$ monadic, with semiabelian $\mathcal{X}$;
3. – There exists $U : \mathcal{A} \to \mathcal{X}$ monadic, with semiabelian $\mathcal{X}$;
   – $F \dashv U$ such that $T = UF$ preserves regular epi and kernel pairs.

Ideally exact categories

Theorem

Given an object A in a semiabelian category, it holds that

$$\mathrm{NSub}(A) \overset{\cong}{\longleftrightarrow} \mathrm{Quot}(A)$$

Theorem

Given an object A in an ideally exact category $\mathcal{A}$, and $U : \mathcal{A} \to \mathcal{X}$ the associated functor. If U is (up to iso) $!^*$, it holds that

$$\mathrm{NSub}\big(U(A)\big) \overset{\cong}{\longleftrightarrow} \mathrm{Quot}(A)$$

Ideally exact categories

Definition

Let $\mathcal{A}$ be an ideally exact category, and $U : \mathcal{A} \to \mathcal{X}$ the associated functor. The lattice $\mathrm{NSub}(U(A))$ consists of U-**ideals** of A.

If U is $!^*$ (up to iso), the U-ideals of A are simply called **ideals**.

Example

R unital ring, $I \lhd R$ an ideal

$$I \hookrightarrow R \xrightarrow{\pi} R/I$$

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Cotopos

Definition

A category $\mathcal{E}$ is said to be an **elementary topos**, if

- It is finitely complete;
- It is cartesian closed;
- It has a subobject classifier.

A **subobject classifier** of $\mathcal{E}$ is a pair (Ω, t), with t mono, such that for any object A and any mono $m : S \rightarrowtail A$, there exists a unique $\chi_S : A \to \Omega$ making the diagram a pullback.

$$\begin{array}{ccc} S & \longrightarrow & 1 \\ {\scriptstyle m}\downarrow & & \downarrow{\scriptstyle t} \\ A & \xrightarrow{\chi_S} & \Omega \end{array}$$

Cotopos

Theorem

A cotopos is an ideally exact category.

Example

$\underline{Set}^{op}$ is a cotopos, thus ideally exact.

Cotopos

$$\left\{ \begin{array}{c} \text{Classes of iso of} \\ \text{inject func in } \underline{Set} \\ \text{with codom } A \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Classes of iso of} \\ \text{cokers in } \underline{Set}^{\bullet} \\ \text{with dom } (\{*\} \sqcup A, *) \end{array} \right\}$$

$$\left[\, f : B \rightarrowtail A \, \right] \quad \longmapsto \quad \left[\, \bar{f} \, \right]$$

$$\left[\, g^{-1}(c) \cap A \rightarrowtail A \, \right] \quad \longleftarrow \quad \left[\, g : (A \sqcup \{*\}, *) \twoheadrightarrow (C, c) \, \right]$$

$$\bar{f} : \big(A \sqcup \{*\}, *\big) \twoheadrightarrow \big((A \setminus f(B)) \sqcup \{*\}, *\big)$$

$$a \mapsto \begin{cases} * & \text{if } a \in f(B) \\ a & \text{otherwise} \end{cases}$$

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